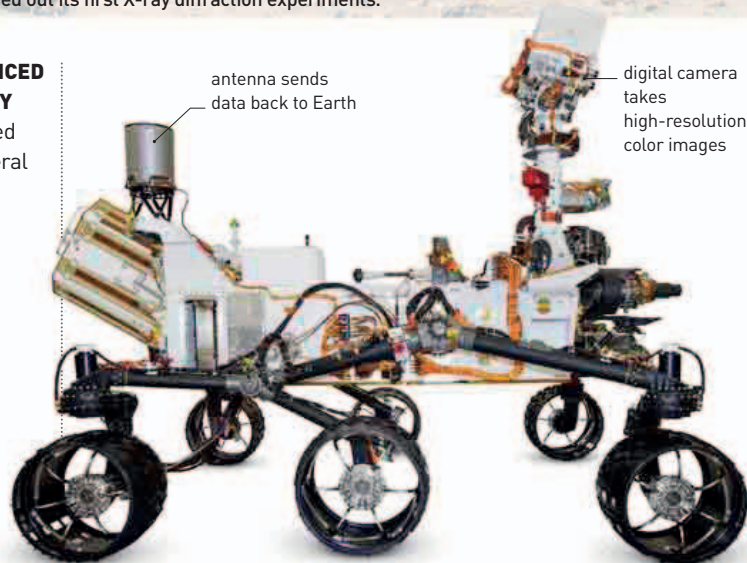


This image shows the "Rocknest," an area of the Martian surface where Curiosity found mysterious bright objects and where it carried out its first X-ray diffraction experiments.

IN MAY 2011, NASA ANNOUNCED RESULTS FROM ITS GRAVITY PROBE B, which was designed to test Albert Einstein's general theory of relativity, originally published in 1916. Since its launch in 2004, the probe had been measuring the curvature of space-time in the vicinity of Earth (see pp.244-45), and the extent to which Earth's rotation drags space-time around with it. The results of both tests provided the **best confirmation to date of Einstein's general theory of relativity.**

In June, a team of geophysicists presented the **first map of the terrain beneath the Antarctic ice sheet.** Part of a long-term project to examine the geology of Antarctica, the map was produced from data provided by several instruments, including ice-penetrating radar. It revealed various geological features in glacial landscapes and also gave valuable information about the icecap's



formation, which began about 30 million years ago. In August, a team of Australian and British scientists found **microfossils in rocks 3.4 billion years old,** pushing back the date of the earliest known life on Earth by a few million years. The primitive cells' metabolisms were based on sulfur rather than oxygen.

The following year, 2012, was a momentous one for science as physicists at CERN announced a

result of experiments at the Large Hadron Collider (LHC, see 2008). The LHC aims to recreate energies and conditions that existed a tiny fraction of a second after the Big Bang (see pp.344-45), when the Universe formed. On July 4, 2012, physicists at CERN announced that they had found **compelling evidence for the existence of the Higgs boson.** This particle is a

mainstay of the Standard Model of particle physics (see 1974) and is associated with the Higgs field. According to a theory developed by British physicist Peter Higgs and others in the 1960s, the Higgs field exists throughout space, and interaction with the Higgs field is **responsible for giving fundamental**

Roving over Mars

NASA's Curiosity rover is about the size of a family car. It carries an array of instruments, including ones designed to detect chemical compounds that could support life.

particles, such as quarks and leptons, **their mass.**

In August 2012, NASA's **Curiosity rover landed in a large crater (the Gale crater) on Mars,** beginning the most comprehensive mission to the planet so far. The landing took place autonomously, during a radio black-out that NASA engineers called "seven minutes of terror." In the last stage of descent, four rockets fired to slow the craft to almost a hover and the rover itself was lowered to the surface on cables to prevent kicking up dust that could damage its instruments.

Soon after landing, the rover began transmitting stunning, **high-resolution panoramas and close-ups from Mars.** As well as cameras, Curiosity also carries a number of instruments to collect and analyze regolith (soil) and rock samples. A laser vaporizes any rock samples of interest and a spectroscope records the spectrum of the light emitted by the vapor to determine the rock's makeup. Other instruments work out the crystal structure of minerals using X-ray diffraction, and an environmental monitoring station records temperature, wind speed, atmospheric pressure, and humidity. By the end of 2012, Curiosity had traveled more than 1,650 ft (500 m) and had **analyzed regolith from more than 30 different locations.**

1,982 lb THE MASS OF CURIOSITY
9.5 ft THE LENGTH OF CURIOSITY
7.2 ft THE HEIGHT OF CURIOSITY

PETER HIGGS (b.1929)

Theoretical physicist Peter Higgs was born in Newcastle-upon-Tyne, UK. In the early 1960s, he developed a theoretical mechanism (which was also proposed by several other physicists at around the same time) to explain why particles have mass. In 1964, he predicted the existence of a particle associated with that mechanism: the Higgs boson.

